

The Barn Effect: A Streaming Algorithm for Pairwise Feature Attribution on Tabular Data

Streaming Estimation of Log-Weighted Pairwise Conditional Means

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Abstract

We present *The Barn Effect*, a two-phase algorithm for per-sample feature attribution on tabular data supporting regression, binary classification, and multi-class classification targets. Conceptually Φ is a *streaming sufficient-statistic summary* of pairwise stratum-wise outcome structure under a chosen quantization $\mathcal{B}$. The first phase discretizes each column into a finite set of labeled bins via frequency-ranked boundaries, building a compact vocabulary of value labels shared across all columns. The second phase accumulates, for every pair of columns and every co-occurring pair of bin labels, a *log-frequency-weighted conditional mean* of the outcome variable, producing a sparse interaction graph Φ as the learned model artifact. Both phases are streaming-compatible, operating on fixed-size chunks without materializing the full dataset. The outcome $Y^{(n)}$ is *parametric-optional*: it may be the raw target label or the predicted output of any trained model (gradient-boosted trees, random forests, neural networks, etc.), making the algorithm applicable both as a direct data analysis tool and as a model-agnostic explainer. At inference, each sample’s bin labels are routed through Φ via edge lookups, producing a per-column attribution score for that row. Averaging these scores across rows yields a global feature importance ranking; reading them per-row yields individual prediction explanation. We provide formal definitions, a convergence result connecting Φ to conditional expectation, and complexity bounds scaling as $O(p^2N)$ in training and $O(p^2)$ per inference sample.

1 Introduction

A recurring challenge in applied machine learning is understanding *which features drive an outcome for a specific sample*, rather than on average across a population. Global feature importance methods—permutation importance, coefficient magnitudes, tree-based impurity scores—assign a single weight to each feature regardless of what the input values actually are. Per-sample attribution methods address this gap but often require a differentiable model [7], a fitted ensemble [9], or a Shapley-value computation whose cost is exponential in the number of features [8].

The Barn Effect proceeds differently: it operates directly on the data by recording, for every pair of feature values that co-occur in a training row, what outcome that co-occurrence is associated with. These pairwise statistics accumulate into a sparse weighted graph Φ —the *interaction graph*—whose entries are log-frequency-weighted conditional means of the outcome variable. At inference, a sample’s bin labels are looked up in Φ , and the resulting edge weights are aggregated per column to produce attribution scores.

Two-phase structure.

1. **Training (graph construction).** Discretize columns into bins; accumulate co-occurrence statistics over the data in chunks; compute Φ as the completed interaction graph.
2. **Inference (edge lookup).** Map each row’s values to their bin labels; look up the edge weight in G_Φ for every column pair; aggregate per column to produce the attribution matrix Δ .

The model artifact is Φ : a sparse matrix of shape $V \times V$, where V is the size of the bin-label vocabulary. Once trained, Φ can be applied to any number of rows without retraining.

Parametric-optional outcome. The outcome $Y^{(n)}$ fed into the accumulator is not required to be the raw data label. It may equally be the scalar output of any trained model $\hat{f}(\mathbf{x}^{(n)})$ —a gradient-boosted score [10], a random forest prediction [9], a neural network logit, or a calibrated probability. When raw labels are used, the Barn Effect describes the data directly. When model outputs are used, it explains the model’s behavior: which feature co-occurrences are associated with high or low predicted scores.

Streaming scalability. A key practical advantage over existing attribution methods is that the Barn Effect never requires the full dataset to be held in memory simultaneously. SHAP [7] requires a background dataset at inference time, and exact Shapley computation is exponential in p ; LIME [8] requires repeated model queries per sample, with neither method natively supporting data streams or datasets that exceed available memory. The Barn Effect’s sufficient statistics $(\mathbf{W}, \mathbf{C})$ grow only with the number of distinct co-occurring bin-label pairs—bounded by $O(p^2 B^2)$ regardless of N —enabling attribution over arbitrarily large or continuously arriving data without retaining the original dataset [12, 13]. Combined with additive accumulation (Section 4), this makes the algorithm applicable to continual and stream-learning settings [12] where static explainability methods cannot be applied incrementally.

Scope

We do not claim experimental benchmarks or comparison against existing attribution methods. We provide definitions, constructions sufficient for implementation, and the mathematical properties of the interaction graph and its derived attribution scores.

2 Preliminaries and notation

Let $[n] := \{1, \dots, n\}$. Scalars are Roman or Greek; vectors are bold lower case (e.g., $\mathbf{v}$); matrices are bold upper case (e.g., $\mathbf{A}$). Element-wise matrix operations are written with $\odot$ (product) and $\oslash$ (division on the nonzero support). For a finite set S , $|S|$ denotes its cardinality.

Dataset. Let $\mathcal{D} = \{(\mathbf{x}^{(n)}, Y^{(n)})\}_{n=1}^N$ be a tabular dataset with p features and scalar outcome $Y^{(n)} \in \mathbb{R}$. We write $X_i^{(n)}$ for the value of feature $i \in [p]$ at sample n .

Definition 1 (Outcome). $Y^{(n)} \in \mathbb{R}$ is a scalar outcome; see §7 for parametric-optional usage.

Column pair set. We require $p \geq 2$.

$$\mathcal{C} := \{(i, j) : 1 \leq i < j \leq p\}, \quad |\mathcal{C}| = \frac{p(p-1)}{2}.$$

Data may arrive in K contiguous chunks $\mathcal{D}_1, \dots, \mathcal{D}_K$ of sizes $n_1, \dots, n_K$ with $\sum_k n_k = N$.

3 Phase 1: Discretization

Raw feature values are mapped to discrete bin labels before accumulation. Discretization limits the vocabulary of value pairs that must be tracked and groups continuous or sparse numerical values into meaningful ranges.

Definition 2 (Bin schema). *Fix a per-column bin count $B_i \in \mathbb{N}$ for $i \in [p]$, denoting the desired number of output bins.*

- **Numerical column i .** Let $r_i := \max_n X_i^{(n)} - \min_n X_i^{(n)}$ be the observed range of column i . Round all observed values to s_i decimal places, where s_i is given by (1). Select the top- B_i most frequently occurring rounded values as boundary candidates; sort them and replace the largest with the observed column maximum $b_{B_i}^{(i)} := \max_n X_i^{(n)}$; prepend the observed minimum as $b_0^{(i)} := \min_n X_i^{(n)}$. This yields $B_i + 1$ boundary values $b_0^{(i)} < b_1^{(i)} < \dots < b_{B_i}^{(i)}$, with $b_1^{(i)}, \dots, b_{B_i-1}^{(i)}$ as the $B_i - 1$ interior split points and the endpoints anchored at the observed data range. The bin label assigned to value x is the interval string $[b_k^{(i)}, b_{k+1}^{(i)})$, where $k = \max\{k' \in \{0, \dots, B_i - 1\} : b_{k'}^{(i)} \leq x\}$. This yields exactly B_i non-empty bins.
- **Categorical column i .** Let $\mathcal{F}_i$ be the B_i most frequently occurring distinct values. Any value outside $\mathcal{F}_i$ maps to the distinguished token ε . The label set is $\mathcal{L}_i = \mathcal{F}_i \cup \{\varepsilon\}$.

The bin schema $\mathcal{B} = \{(\mathcal{L}_i, \ell^{(i)})\}_{i=1}^p$ collects the label sets and the maps $\ell^{(i)} : \text{dom}(X_i) \rightarrow \mathcal{L}_i$.

For numerical columns, the integer s_i adjusts rounding coarseness from the observed range:

$$s_i := \max(0, 4 - \lfloor \log_{10}(r_i) \rfloor), \quad (1)$$

so wider r_i yields coarser rounding and narrower r_i finer rounding (e.g., $r_i \approx 1$ gives $s_i = 4$; $r_i \approx 1000$ gives $s_i = 1$).

Missing inputs before binning. The maps $\ell^{(i)}$ are applied only after a practitioner-chosen *missing-input convention* maps raw cells to inputs in $\text{dom}(X_i)$. §7 fixes one default pipeline (numerical NaN $\rightarrow$ 0, categorical missing $\rightarrow \varepsilon$ when rare, outcome NaN $\rightarrow$ 0 for accumulation). Histograms used for frequencies, endpoints min / max, and hence r_i and $\mathcal{B}_N$, are computed from these *effective* values, so systematic missingness can concentrate mass on imputed atoms and couples $\mathcal{B}_N$ to the missing-data rule (potentially analogous to MAR/MNAR issues in exploratory analysis).

Figure 1 illustrates numerical binning with $B = 4$: the top-4 most frequent values are selected as boundary candidates; the largest is replaced by the column maximum, leaving three interior split points and yielding exactly $B = 4$ bins. Figure 2 illustrates categorical selection with $B = 4$: the four most frequent categories are retained; all others collapse to ε .

Definition 3 (Bin-label vocabulary). *Let $\mathcal{V} := \bigcup_{i=1}^p \mathcal{L}_i$ be the set of all bin labels produced by $\mathcal{B}$, with $V := |\mathcal{V}|$. Identify $\mathcal{V}$ with $[V]$ via a fixed ordering established during discretization. From this point forward, bin labels are integers in $[V]$; in particular, $\ell^{(i)}(X_i^{(n)}) \in [V]$ for all i and n .*

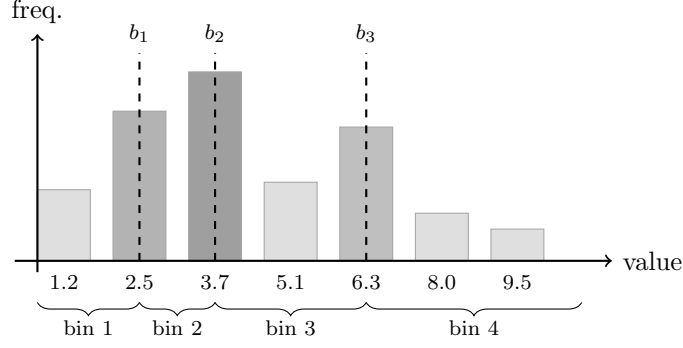


Figure 1: Numerical bin schema with $B = 4$ bins. The top-4 most frequent rounded values are the boundary candidates (darker bars); the largest is replaced by the observed column maximum, leaving three interior split points b_1, b_2, b_3 (dashed lines) and yielding $B = 4$ bins.

Remark 1 (Column-specific labels). *Each label set $\mathcal{L}_i$ is disjoint from $\mathcal{L}_j$ for $i \neq j$: the encoding assigns distinct integers to labels from different columns even when their raw values are identical. As a result, any $u \in \mathcal{L}_i$ and $v \in \mathcal{L}_j$ with $i \neq j$ are always distinct integers in $[V]$, and the diagonal $\Phi_{u,u}$ is never populated by cross-column pairs.*

Remark 2 (Streaming compatibility of Phase 1). *The bin schema $\mathcal{B}$ can be finalized in a single forward pass without reloading any chunk: accumulate per-column rounded-value counts, min, and max across chunks; rank frequencies and assign boundaries once after the final chunk.*

4 Phase 2: Pairwise accumulation and graph construction

After discretization, all feature values are integers in $[V]$. Phase 2 processes every column pair, accumulates sufficient statistics for each co-occurring value pair, and finalizes the interaction graph Φ .

Definition 4 (Value pair). *For sample n and column pair $(i, j) \in \mathcal{C}$, the value pair is*

$$(u, v) := (\ell^{(i)}(X_i^{(n)}), \ell^{(j)}(X_j^{(n)})) \in [V]^2.$$

Definition 5 (Streaming accumulator). *Maintain two sparse matrices $\mathbf{W}, \mathbf{C} \in \mathbb{R}^{V \times V}$, initialized to $\mathbf{0}$. For each chunk $\mathcal{D}_k$, column pair $(i, j) \in \mathcal{C}$ (with $i < j$), define the per-chunk increments*

$$\Delta W_{uv}^{(k)} := \sum_{\substack{n \in \mathcal{D}_k: \\ \ell^{(i)}(X_i^{(n)})=u, \ell^{(j)}(X_j^{(n)})=v}} Y^{(n)}, \quad (2)$$

$$\Delta C_{uv}^{(k)} := |\{n \in \mathcal{D}_k : \ell^{(i)}(X_i^{(n)}) = u, \ell^{(j)}(X_j^{(n)}) = v\}|. \quad (3)$$

Pairs (u, v) absent from chunk $\mathcal{D}_k$ contribute $\Delta W_{uv}^{(k)} = \Delta C_{uv}^{(k)} = 0$. After all K chunks the final values are

$$W_{uv} := \sum_{k=1}^K \Delta W_{uv}^{(k)}, \quad C_{uv} := \sum_{k=1}^K \Delta C_{uv}^{(k)},$$

where (u, v) is always keyed in the canonical column order $i < j$ imposed by $\mathcal{C}$.

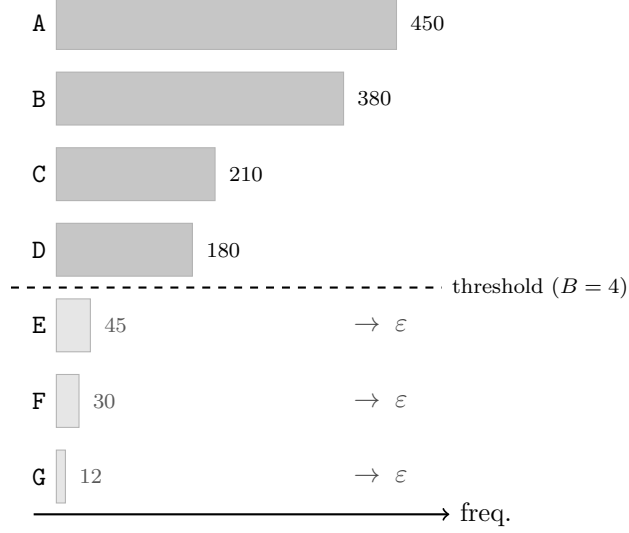


Figure 2: Categorical bin schema with $B = 4$. The four most frequent categories (A–D) are retained; the remaining three fall below the frequency threshold and are mapped to the distinguished token ε .

Remark 3 (Additive accumulation). *Because the increments (2)–(3) are additive and the chunks $\mathcal{D}_1, \dots, \mathcal{D}_K$ are disjoint, W_{uv} and C_{uv} depend only on the full data $\mathcal{D}$, not on how $\mathcal{D}$ is partitioned into chunks. Consequently, $(\mathbf{W}, \mathbf{C})$ are sufficient statistics for Φ : new data may be incorporated by resuming accumulation from the existing totals, with no prior samples retained.*

Definition 6 (Interaction graph). *The interaction graph is the sparse matrix $\Phi \in \mathbb{R}^{V \times V}$ with entries*

$$\Phi_{uv} := \begin{cases} 0 & \text{if } C_{uv} \in \{0, 1\}, \\ \frac{W_{uv}}{C_{uv}} \cdot \log_{10}(C_{uv}) & \text{otherwise,} \end{cases} \quad (4)$$

equivalently $\Phi = (\mathbf{W} \odot \mathbf{C}) \odot \log_{10}(\mathbf{C})$ on the support $\{(u, v) : C_{uv} > 1\}$.

The matrix Φ is the weighted adjacency matrix of the *interaction graph*

$$G_\Phi = (\mathcal{V}, E_\Phi, w_\Phi),$$

where $\mathcal{V} = [V]$ is the bin-label vocabulary, $E_\Phi = \{(u, v) : \Phi_{uv} \neq 0\}$ is the edge set, and $w_\Phi(u, v) = \Phi_{uv}$ is the edge weight function. We use Φ and G_Φ interchangeably: Φ for matrix operations, G_Φ for graph language. The $C_{uv} = 1$ boundary case falls out naturally: $\log_{10}(1) = 0$, so a single observation contributes zero; together with $C_{uv} = 0$ this yields $\Phi_{uv} = 0$ for both absent and singleton pairs.

Remark 4 (Undirected edges). *The co-occurrence event “column i at bin u , column j at bin v ” is identical to “column j at bin v , column i at bin u ”: there is no inherent direction. The accumulator stores each event under a single canonical key determined by column order $i < j$; inference recovers the correct edge weight by applying the same ordering (see Section 7). The interaction graph G_Φ is therefore undirected.*

5 Inference: attribution from the interaction graph

Definition 7 (Per-sample pairwise score). *For sample n and column pair $(i, j) \in \mathcal{C}$, the pairwise score is*

$$\psi_{ij}^{(n)} := \Phi_{\ell^{(i)}(X_i^{(n)}), \ell^{(j)}(X_j^{(n)})}. \quad (5)$$

This is a single entry lookup in G_Φ : given the bin labels for columns i and j at sample n , retrieve the edge weight Φ_{uv} . If $(u, v) \notin E_\Phi$, then $\psi_{ij}^{(n)} = 0$.

Definition 8 (Column attribution score $A_c^{(n)}$). *Let $d_c^{(n)} := |\{j \neq c : \psi_{cj}^{(n)} \neq 0\}|$ be the number of column partners with a nonzero pairwise score for column c in sample n . The column attribution score is*

$$A_c^{(n)} := \begin{cases} 0 & d_c^{(n)} = 0, \\ \frac{1}{d_c^{(n)}} \sum_{\substack{j=1 \\ j \neq c}}^p \psi_{cj}^{(n)} & d_c^{(n)} > 0. \end{cases} \quad (6)$$

Normalizing by $d_c^{(n)}$ rather than $p - 1$ excludes value pairs absent from the training data from the average, rather than treating their absence as a zero signal.

Remark 5 (Shared pairwise edges). *Each entry Φ_{uv} of the interaction graph contributes to the attribution of both the column that produced label u and the column that produced label v . The same edge weight appears in $A_c^{(n)}$ for column c and in $A_{c'}^{(n)}$ for any other column c' that shares an encoded bin in that pair. Unlike Shapley-based methods, the Barn Effect makes no attempt to allocate the joint effect of a co-occurrence to one side: both columns “see” the same signal, and the user interprets their attributions as correlated evidence rather than a partition of a fixed budget.*

Definition 9 (Barn attribution). *Let $\bar{Y} := N^{-1} \sum_{n=1}^N Y^{(n)}$ be the global outcome mean. The Barn attribution for column c at sample n is*

$$\delta_c^{(n)} := A_c^{(n)} - \bar{Y}. \quad (7)$$

A positive value of $\delta_c^{(n)}$ indicates that column c ’s pairwise context in row n is associated with outcomes above the global mean; a negative value indicates the opposite.

Remark 6 (No efficiency axiom). *Unlike SHAP, the Barn Effect does not satisfy the efficiency axiom: there is no guarantee that*

$$\sum_{c=1}^p \delta_c^{(n)} = Y^{(n)} - \bar{Y}.$$

The attributions $\delta_c^{(n)}$ are not a decomposition of the outcome gap into per-feature shares; they are estimated descriptions of how strongly each feature’s pairwise context is associated with a deviation from the baseline. Two features may both exhibit positive attributions without their sum equaling the observed deviation, because each $\delta_c^{(n)}$ is computed from an independent average over pairwise interactions, not from an additive partition of a fixed budget.

The full output of inference is the *attribution matrix* $\Delta \in \mathbb{R}^{N \times p}$ with $\Delta_{nc} = \delta_c^{(n)}$.

Definition 10 (Global feature importance). *The global importance of feature c is the mean Barn attribution across all samples:*

$$\bar{\delta}_c := \frac{1}{N} \sum_{n=1}^N \delta_c^{(n)}. \quad (8)$$

6 Properties

Remark 7 (Chunk-additive accumulation). *Because increments (2)–(3) are additive over disjoint chunks, for every (u, v) the totals W_{uv} and C_{uv} agree with those obtained from a single uninterrupted pass over $\mathcal{D}$. Chunking is a streaming implementation detail; it does not change the defined statistics.*

Proposition 1 (Complexity). *Let N be the total sample count, p the feature count, and $B = \max_i B_i$ the maximum bin depth.*

1. **Training.** *Processing all K chunks requires $O(p^2 N)$ scalar operations. Space for $\mathbf{W}$ and $\mathbf{C}$ is $O(p^2 B^2)$ entries in the worst case.*
2. **Inference.** *Computing Δ for a batch of M samples requires $O(p^2 M)$ hash-map lookups, each $O(1)$ expected.*

Proof sketch. $|\mathcal{C}| = p(p-1)/2$. For each chunk of size n_k , processing one column pair requires stacking n_k value pairs and computing unique counts in $O(n_k)$ via vectorized sort or hash; summing over all pairs and chunks gives $O(p^2 N)$. Each column pair contributes at most B^2 distinct encoded pairs, so $|\mathcal{C}| \cdot B^2 = O(p^2 B^2)$ accumulator entries in total. At inference, $A_c^{(n)}$ requires $p-1$ lookups per column over p columns: $O(p^2)$ per sample. $\square$

Assumption 1 (Fixed quantization for asymptotics). *Throughout Section 6, and in particular Proposition 2, the bin schema $\mathcal{B}$ and the quantization maps $\ell^{(i)}$ are nonrandom measurable functions of the covariates fixed before taking limits in N : the strata $\{\ell^{(i)}(X_i) = u, \ell^{(j)}(X_j) = v\}$ and the masses π in Proposition 2 are thus well-defined under the data-generating measure and do not vary with sample size. Section 3 instead prescribes a sample-constructed schema $\mathcal{B}_N$ —random and N -dependent. Practitioners configure the depth B_i and the frequency-based construction (see also §7); one operational way to approximate Assumption 1 is to freeze $\mathcal{B}$ from a pilot or hold-out slice, then accumulate $\mathbf{W}, \mathbf{C}$ on fresh data drawn from the same regime. Extending Proposition 2 to jointly random $(\mathcal{B}_N, \Phi)$ requires additional analysis; we do not pursue it here.*

Proposition 2 (Convergence of Φ to scaled conditional expectation). *Assume Assumption 1. Suppose $\{(\mathbf{x}^{(n)}, Y^{(n)})\}$ are i.i.d. with $\mathbb{E}[|Y|] < \infty$. Fix $(i, j) \in \mathcal{C}$ and $(u, v) \in [V]^2$ with $\pi := P(\ell^{(i)}(X_i) = u, \ell^{(j)}(X_j) = v) > 0$. Then*

$$\frac{\Phi_{uv}}{\log_{10}(N\pi)} \xrightarrow{\text{a.s.}} \mathbb{E}[Y \mid \ell^{(i)}(X_i) = u, \ell^{(j)}(X_j) = v] \quad \text{as } N \rightarrow \infty. \quad (9)$$

Proof sketch. By the strong law of large numbers, $C_{uv}/N \xrightarrow{\text{a.s.}} \pi$. Since $\log_{10}$ is continuous on $(0, \infty)$ and $\pi > 0$, the continuous mapping theorem gives $\log_{10}(C_{uv}/N) \xrightarrow{\text{a.s.}} \log_{10}(\pi)$, and therefore

$$\log_{10}(C_{uv}) = \log_{10}(N) + \log_{10}(C_{uv}/N) \xrightarrow{\text{a.s.}} \log_{10}(N) + \log_{10}(\pi) = \log_{10}(N\pi).$$

Separately, the subsequence of samples with the specified bin labels grows without bound (since $\pi > 0$), so $W_{uv}/C_{uv} \xrightarrow{\text{a.s.}} \mathbb{E}[Y \mid \ell^{(i)}(X_i) = u, \ell^{(j)}(X_j) = v]$ by the strong law applied to that subsequence. Multiplying and dividing by $\log_{10}(N\pi)$ yields (9). $\square$

Remark 8. *Proposition 2 shows Φ_{uv} is a consistent, frequency-scaled estimator of the conditional mean: entries for frequently co-occurring pairs (large C_{uv}) receive larger absolute weights, reflecting greater statistical confidence. Since $\Phi_{uv} = (W_{uv}/C_{uv}) \cdot \log_{10}(C_{uv})$, dividing Φ_{uv} by $\log_{10}(C_{uv})$ recovers the sample conditional mean W_{uv}/C_{uv} , which is a consistent estimator of $\mathbb{E}[Y \mid u, v]$ by the strong law of large numbers. In practice the raw Φ_{uv} values are preferred for attribution because they naturally downweight sparse evidence.*

Remark 9 (Column attributions and random support). *Under Assumption 1, Proposition 2 yields almost-sure stabilization of Φ_{uv} on every fixed cell (u, v) with $\pi > 0$. For a fixed row n and column c , each nonzero $\psi_{cj}^{(n)}$ is an edge lookup into this limit graph; however, $A_c^{(n)}$ averages only over partners j with $\psi_{cj}^{(n)} \neq 0$, with denominator $d_c^{(n)}$ random and data-dependent. No single deterministic limit of the form $A_c^{(n)} \rightarrow (1/d_c) \sum_j(\dots)$ is claimed: aggregating a random subset of dependent edge terms under a random $d_c^{(n)}$ is not summarized by a fixed- d_c asymptotic identity. The interpretive role of $d_c^{(n)}$ and sparse Φ support is discussed in Remark 10.*

Remark 10 (Support and identifiability). *An absent or zero-weight entry Φ_{uv} must not be read as confirming $\mathbb{E}[Y \mid u, v] = 0$. If $C_{uv} = 0$, the stratum (u, v) is unseen in training; if $C_{uv} = 1$, (4) sets $\Phi_{uv} = 0$, suppressing unreliable log-scaled weight rather than asserting a negligible effect. Classic categorical-data analysis warns that sparse and empty contingency cells carry weak asymptotic justification for inferring cell-specific quantities without auxiliary assumptions [1]; similarly, conditional means on unobserved or barely observed strata are typically only partially identifiable alongside many compatible values under the empirical distribution [6].*

The denominator $d_c^{(n)}$ in Definition 8 is therefore deliberate: $A_c^{(n)}$ averages $\psi_{cj}^{(n)}$ only over partners j for which the lookup returns nonzero interaction mass at inference, i.e. co-occurrence patterns that accrued enough training evidence (beyond singleton pairs) under the induced bins. Thus $d_c^{(n)}$ encodes “revealed” pairwise context at the current training snapshot relative to $p - 1$ hypotheticals, matching a descriptive attribution goal rather than a complete allocation of hypothetical counterfactuals.

Finally, distinguish structural from transient sparsity. In Proposition 2’s setting— $\mathcal{B}$ held fixed while $N \rightarrow \infty$, as in Assumption 1—if $\pi_{uv} > 0$ for a cell (u, v) , then $C_{uv} \rightarrow \infty$ almost surely under i.i.d. sampling and, once $C_{uv} > 1$, the construction (4) admits a nontrivial Φ_{uv} . Persistent emptiness as $N \rightarrow \infty$ within that asymptotic regime is instead consistent with $\pi_{uv} = 0$: the algorithm intentionally withholds extrapolated signal for strata that sampling never activates, separating structural non-support from small- N holes that finite samples alone cannot resolve; Assumption 1 isolates limits from empirical $\mathcal{B}_N$.

7 Implementation

The definitions in Sections 3–5 specify what the algorithm computes. This section describes how those computations are realized. Three design principles connect the math to the code: bin labels are column-prefixed integer keys ensuring vocabulary disjointness (Remark 1); $\mathbf{W}$ and $\mathbf{C}$ are stored as a single hash map rather than a dense $V \times V$ matrix, realizing the sparse edge set E_Φ in $O(p^2 B^2)$ space; and inference is a sequence of $O(1)$ hash-map lookups, each keyed by the canonical column order established during training (Remark 4).

Parametric-optional outcome. The practitioner fixes $Y^{(n)} \in \mathbb{R}$ per instance of the accumulator: raw regression targets (possibly standardized); binary labels $\{0, 1\}$ or predicted positive-class probability $\hat{P}(\text{class} = 1 \mid \mathbf{x}^{(n)})$; multi-class setups by running K independent accumulators with $Y^{(n)} = \hat{P}(\text{class} = k \mid \mathbf{x}^{(n)})$ for each k ; or predictions $\hat{f}(\mathbf{x}^{(n)})$ from any trained model $\hat{f}$ to explain model behavior rather than the raw label distribution. No further distributional assumption is required beyond measurability of the chosen scalar.

Phase 1: Vocabulary construction. The bin schema $\mathcal{B}$ is built in a single forward pass with no chunk reloaded (Remark 2). Rounding uses s_i from (1); for each numerical column i , maintain running rounded-value counts, column minimum, and column maximum across chunks; after the final chunk, select the top- B_i most frequent rounded values as boundary candidates, replace the largest with the observed maximum, and prepend the minimum to produce $B_i + 1$ boundary values defining B_i bins. For each categorical column i , merge per-chunk category counts and retain the top- B_i categories; all others map to ε . Each bin label string is prefixed with its column index before insertion into the global string-to-integer map (e.g., interval $[1.2, 3.4)$ in column 2 becomes `col.2:[1.2,3.4)`), guaranteeing that identical raw values in different columns receive distinct integers in $\mathcal{V}$.

Missing values. This implements the Phase 1 missing-input conventions in §3: before bin assignment, numerical NaN is replaced with zero; categorical NaN is treated as the string `empty`, mapping to ε if infrequent; outcome NaN is replaced with zero, contributing zero to W_{uv} for every pair in that row. These rules apply before any accumulation; they embody one choice of measurable missingness map feeding $\ell^{(i)}$.

Phase 2: Accumulating co-occurrence statistics. The sparse matrices $\mathbf{W}$ and $\mathbf{C}$ are realized as a single hash map from canonical integer pairs (u, v) to floating-point values, where $u = \ell^{(i)}(X_i^{(n)})$ and $v = \ell^{(j)}(X_j^{(n)})$ with $i < j$ as ordered in $\mathcal{C}$. For each chunk $\mathcal{D}_k$ and each column pair $(i, j) \in \mathcal{C}$, aggregate co-occurrences within the chunk: for every (u, v) realized by at least one row in $\mathcal{D}_k$, update $W_{uv} += \sum_{\{n \in \mathcal{D}_k : \ell^{(i)}=u, \ell^{(j)}=v\}} Y^{(n)}$ and $C_{uv} += |\{n \in \mathcal{D}_k : \ell^{(i)} = u, \ell^{(j)} = v\}|$. Equivalently, scan each row of $\mathcal{D}_k$ and update the hash map once per column pair. Concurrently, a running sum $S += \sum_{n \in \mathcal{D}_k} Y^{(n)}$ and count $M += n_k$ accumulate the global mean; $\bar{Y} = S/M$ is finalized after the last chunk. Because all updates are additive (Remark 3), the algorithm can be extended to new data at any point by resuming from the existing $(\mathbf{W}, \mathbf{C})$ state and recomputing Φ .

Computing Φ . After all chunks, apply (4) to each entry with $C_{uv} > 1$; discard entries with $C_{uv} \leq 1$. The result is the trained artifact: a sparse hash map from (u, v) pairs to edge weights. The log factor gives diminishing returns to frequency: a pair with conditional mean $\hat{\mu} = 0.9$ observed $C = 3$ times scores $0.9 \times \log_{10}(3) \approx 0.43$, while a pair with $\hat{\mu} = 0.5$ and $C = 100$ scores $0.5 \times 2.0 = 1.00$, so a frequently observed moderate signal outweighs a sparsely observed strong one.

Inference. For each row n : (i) map each $X_i^{(n)}$ to $\ell^{(i)}(X_i^{(n)}) \in [V]$ via $\mathcal{B}$; (ii) for each pair $(i, j) \in \mathcal{C}$, look up Φ_{uv} in the hash map using the canonical key $(\ell^{(i)}, \ell^{(j)})$ — absent keys return 0; (iii) for each column c , recover its scores by looking up $(\ell^{(c)}, \ell^{(j)})$ when $c < j$ and $(\ell^{(j)}, \ell^{(c)})$ when $c > j$; (iv) count the nonzero results as $d_c^{(n)}$ and compute $\delta_c^{(n)} = (d_c^{(n)})^{-1} \sum_{j \neq c} \psi_{cj}^{(n)} - \bar{Y}$, or $-\bar{Y}$ if $d_c^{(n)} = 0$. The full $N \times p$ matrix Δ requires no model access and no background dataset; Φ is queried but not modified.

Row-level and aggregate use. The attribution matrix Δ supports two uses from a single trained Φ . Row-level: $\delta_c^{(n)}$ measures how column c ’s pairwise context for row n deviates from the global baseline — two rows identical in column c but differing in other columns receive different δ_c scores, because $\psi_{cj}^{(n)}$ depends on all columns’ bin labels. Aggregate: $\bar{\delta}_c = N^{-1} \sum_n \delta_c^{(n)}$ ranks features by their consistent association with above- or below-average outcomes across the dataset.

Memory and chunk size. The hash map contains at most $O(p^2 B^2)$ distinct (u, v) keys (Proposition 1), bounded independently of N and V^2 . Chunk size governs peak working memory and I/O granularity while scanning $\mathcal{D}$; because accumulation is additive (Remark 7), the trained Φ does not depend on how rows are grouped into chunks.

8 Related Work

AI Explainability. We position the algorithm relative to a few well-known lines of work the reader may already have in mind. SHAP [7] and LIME [8] are model-querying, local attributions; the introduction already contrasts their cost and data requirements with a streaming sufficient-statistic summary. Partial dependence [11] and ICE curves [5] trace how a *fitted* prediction surface moves with one or two features (population average vs. per row); the Barn Effect instead assigns per-row scores $\delta_c^{(n)}$ from discrete co-occurrence–outcome accumulators, without accessing a model except optionally through the scalar label $Y^{(n)}$.

Association rule mining. Phase 2 is thematically related to frequent itemset mining [2]: both build statistics from co-occurrences of discrete items. Where association-rule methods typically threshold binary support to enumerate candidate rules, the Barn Effect retains, for every observed bin-label pair (u, v) , both the co-occurrence count C_{uv} and the outcome-weighted sum W_{uv} , so that Φ_{uv} targets a supervised conditional mean rather than a boolean presence score. Analogously, NLP pipelines often apply log damping to frequencies (TF-IDF [3], GloVe’s log-smoothed co-occurrences [4]); here $\log_{10}(C_{uv})$ plays a similar damping role but $\mathbf{W}$ is driven by supervised outcomes rather than embedding objectives.

Streaming. The sufficient statistics $(\mathbf{W}, \mathbf{C})$ are additive over rows: bounded storage $O(p^2 B^2)$ for fixed (p, B) and stream-friendly updates [12, 13] without revisiting past observations.

9 Discussion

Parametric-optional design. Using raw labels $Y^{(n)}$ produces a data-driven attribution: Φ encodes which feature co-occurrences are associated with high or low outcome values in the training data. Using predicted model outputs $\hat{f}(\mathbf{x}^{(n)})$ shifts the interpretation: Φ now encodes which co-occurrences drive the model’s predictions, enabling model-agnostic explanation without access to the model’s internals—only its scalar output is required. Both modes produce the same format of Δ ; the difference is entirely in what $Y^{(n)}$ represents.

Chunking and stationary extensions. Additive accumulation (Remark 7) makes the interaction graph invariant to chunk boundaries under a stationary data stream. When the regime is non-stationary (e.g., temporal drift), a constant Φ trained over long horizons mixes regimes; emphasizing recent observations would require deliberate modifications such as decay weights or sliding windows rather than reshaping chunks alone.

Correlated features. When two features are highly correlated, their bin labels tend to co-occur with similar partner values across the dataset. As a result, their pairwise scores $\psi_{cj}^{(n)}$ will be similar and their column attributions $\delta_c^{(n)}$ will reflect those shared patterns rather than isolating which of the two drives the outcome. This is the expected and honest behavior: the algorithm records

what the data actually shows, not a causal partition of credit. High correlation does not distort the attributions in the sense of introducing false signal—it correctly reports that both features are associated with the same outcome context. Distinguishing causation from correlation in such cases requires additional analysis (e.g., controlled interventions or domain knowledge) that lies outside the scope of a descriptive attribution method.

Domain applicability. The two-phase structure generalizes beyond tabular data by substituting the encoding step while leaving the accumulator, inference, and convergence results unchanged. Natural instantiations include sequential token pairs in text, spatially adjacent pixel pairs in images, and cosine similarity scores as outcomes in retrieval settings. In each case, Phase 1 reduces the domain to a finite integer vocabulary; Phase 2 and inference then proceed identically to the tabular formulation. A detailed treatment of these extensions, including multi-modal embeddings via cluster assignment, appears in the companion note `barn_effect_extensions.tex`.

Speculative / heuristic: parametric gap graph. This subsection is *not* an evaluated diagnostic: no calibration, benchmarks, or hypothesis tests are claimed. Consider training Φ_{data} from raw labels $Y^{(n)}$ and Φ_{model} from model outputs $\hat{f}(\mathbf{x}^{(n)})$ on the same $\mathcal{B}$; on their shared edge support define

$$\mathbf{G} := \Phi_{\text{data}} - \Phi_{\text{model}}, \quad (10)$$

informally termed a *gap graph*. Large $|G_{uv}|$ is sometimes read as divergence between empirical co-outcome associations and model-implied associations for that discrete co-occurrence pattern; interpretations of sign (over- vs. underprediction) remain tentative without validated links to realized errors.

The subtraction is only meaningful when both graphs use outcomes on comparable scales—e.g. $Y^{(n)}$ for Φ_{data} and $\hat{f}(\mathbf{x}^{(n)})$ for Φ_{model} drawn from the same rows. Treat $|G_{uv}|$ as exploratory only: without statistical validation or domain-specific evaluation, it should not be advanced as formal model criticism.

Further extensions. Natural directions include: (i) *Temporal weighting*: replace $Y^{(n)}$ with $w^{(n)}Y^{(n)}$ where $w^{(n)} = \exp(-\lambda(t_{\text{max}} - t^{(n)}))$ to emphasize recent observations; (ii) *Third-order interactions*: extend $\mathcal{C}$ to triples (i, j, k) at $O(p^3N)$ training cost; (iii) *Normalized scores*: divide Φ_{uv} by $\log_{10}(C_{uv})$ at inference to recover conditional mean estimates in the original outcome scale; and (iv) *Sketching*: approximate heavy co-occurrence structure (e.g., via count-min sketches) when exact $O(p^2B^2)$ storage is prohibitive.

10 Conclusion

The Barn Effect is a two-phase algorithm for per-sample feature attribution on tabular data. Phase 1 discretizes each column into a finite bin vocabulary using frequency-ranked boundaries. Phase 2 accumulates, for every co-occurring pair of bin labels across all column combinations, a log-frequency-weighted conditional mean of the outcome into a sparse interaction graph Φ . Both phases operate in fixed memory over data chunks, making the algorithm applicable to datasets that exceed available memory and to continually arriving data streams. At inference, each row’s bin labels are routed through Φ via edge lookups, producing a per-column attribution score that measures how much each feature’s pairwise context deviates from the global outcome baseline.

The algorithm requires no model access, no gradient computation, and no background dataset at inference time. The outcome $Y^{(n)}$ is parametric-optional: raw labels produce a data-driven

attribution; model outputs produce a model-agnostic explanation. The interaction graph Φ is a reusable artifact—trained once, queried arbitrarily many times—whose entries converge almost surely to scaled conditional expectations of the outcome under Assumption 1 (Proposition 2).

Future directions include controlled empirical benchmarks against SHAP and ICE on standard attribution tasks, theoretical characterization of the empirical bin schema $\mathcal{B}_N$ in Proposition 2, and extensions to higher-order interactions and non-tabular modalities as developed in the companion note `barn_effect_extensions.tex`.

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